An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss forces internal backpropagation gradients to align directly with true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

The digital transformation of tertiary healthcare facilities has produced cyber-physical ecosystems where uninterrupted electrical supply is not merely an op-

erational convenience but a patient-safety imperative. Modern smart hospitals operate localized microgrids that integrate utility feeders, rooftop photovoltaic arrays, battery energy storage, and emergency diesel generators to power life-critical loads: mechanical ventilators and extracorporeal membrane oxygenation circuits in intensive-care units, robotic surgical suites requiring sub-second voltage stability, mRNA vaccine cold-chains maintained at $-80°$C, and Internet-of-Medical-Things (IoMT) edge gateways linking bedside monitors to electronic health record servers [5,19]. An undetected voltage collapse or spurious load-shedding command initiated by a cyberattack on this infrastructure does not merely incur financial loss—it threatens lives.

Supervisory Control and Data Acquisition (SCADA) systems stream bus-level measurements—voltage magnitude V, phase angle θ, active power P, reactive power Q—from Remote Terminal Units (RTUs) into Weighted Least Squares (WLS) state estimators several times per minute. The classical bad-data detector applies a χ^2 residual threshold to flag corrupted telemetry. Liu et al. [15] demonstrated that an adversary with partial knowledge of the network admittance matrix can construct a stealthy attack vector $a = Hc$, where H is the measurement Jacobian and c is the targeted state bias, such that the WLS residual remains unchanged. The injected corruption cancels inside the χ^2 test, leaving the state estimator converging to a biased solution while hospital operators remain unaware [13].

Bypassing deterministic circuit laws, machine learning detectors initially gained operational traction by projecting raw telemetry directly onto attack flags [12,3]. Yet their architectural flaw is glaring: rigid input vectors assume immutable grid geometry. When line maintenance or breaker actions re-route power flows, fixed sensor indices decouple from active connections, triggering widespread false alarms. Continuous online retraining after switching events remains fundamentally infeasible across mission-critical clinical substations.

Graph Neural Networks (GNNs) circumvent rigid geometry by treating substations as nodes and transmission corridors as dynamic edges [21]. Topological line trips merely contract the adjacency set without disrupting local node feature semantics. Boyaci et al. [2] corroborated this structural invariance through ARMA spectral filters, and Takiddin et al. [22] validated transferability across non-uniform grid layouts. Even so, typical isotropic message passers suffer from an inherent drawback: uniform neighbor pooling blurs localized signal variance, making an attacked substation indistinguishable from adjacent uncompromised nodes in tightly coupled electrical clusters.

Beyond message passing, practical deployment faces two stubborn hurdles. First stands the explainability–training gap: iterative explainers (SHAP, LIME) overrun tight SCADA cycles, while ordinary post-hoc attributions leave model weights untouched [27]. Second, passive alerting cannot safeguard patient care; alarms must autonomously trigger sensor quarantine and physics-based measurement reconstruction so medical loads stay energized during field inspections.

1.1 Contributions

To overcome these structural limitations, our work delivers four key innovations:

- **Physics-Anchored State Synchronization:** Rather than treating the digital twin as an abstract concept, we derive a recursive state-space update equipped with steady-state tracking gain K_{sync} and an adaptive baseline filter to anchor physical-to-virtual alignment.
- **Admittance-Conditioned Graph Message Passing:** We embed branch admittance parameters $Y_{uv} = G_{uv} + jB_{uv}$ directly into the GraphSAGE aggregation kernel, integrating circuit physics into spatial neural embeddings.
- **Attribution-Guided Loss Formulation:** We design an auxiliary gradient-alignment penalty $\mathcal{L}_{\text{align}}$ that steers attribution mass onto compromised bus coordinates during backpropagation, yielding a 5.52 pp gain in localization F1.
- **Closed-Loop Autonomous Telemetry Reconstruction:** We establish an automated quarantine protocol that interpolates missing measurements from healthy electrical neighbors, cutting post-attack WLS estimation error by 99.97% in under 2 ms.

The remainder of this paper examines related literature (Section 2), derives the non-linear AC threat model (Section 3), formulates the cognitive digital twin and admittance GraphSAGE backbone (Section 4), details axiomatic XAI and closed-loop healing (Section 5), presents empirical benchmarks across topologies and scales (Sections 6–7), and discusses clinical deployments (Section 8).

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Integrating hospital IT with industrial OT expands the facility attack surface into clinical substations [11,5]. Corrupting SCADA telemetry risks misleading automated dispatchers into cutting power to surgical suites or intensive care wards. In surveying medical cyber-physical threats, Sun et al. [19] underscored substation spoofing as especially destructive. While Chockalingam et al. [4] developed Bayesian SCADA risk models and Pirbhulal et al. [17] proposed IoT healthcare defense frameworks, existing approaches treat telemetry as isolated scalar streams, neglecting the physical admittance topology governing electrical flows.

2.2 False Data Injection Attacks in Cyber-Physical Energy Systems

Liu et al. [15] proved that adversaries injecting $a = Hc$ manipulate state estimates while bypassing WLS residual checks. This stealth vulnerability spans nonlinear AC regimes, unobserved substations, and decentralized microgrids [13,6], with parallel exploits shown in DC systems [3,18]. Defensive schemes migrated

from shallow learning to deep neural networks [12] and recurrent temporal filters [14]. Yet these methods retain a critical blind spot: binding inputs to fixed coordinate vectors causes detection to collapse when contactors or breakers alter the active circuit path.

2.3 Graph Neural Networks for Grid Security

Graph models overcome this rigidity by anchoring electrical states to vertices and routing messages across physical transmission branches [21]. Early implementations used spectral convolutions [10] or graph attention [25]. To curb oversmoothing in large networks, Hamilton et al. [8] introduced GraphSAGE, concatenating self-embeddings with pooled neighborhoods. In power systems, Boyaci et al. [2] attained 96.25% detection F1 via ARMA wavelets, while Takiddin et al. [22] and Yue and Qu [28] showed GNNs navigate irregular layouts better than dense classifiers. Luo et al. [16] examined federated models for privacy. Even so, existing GNNs omit branch admittance during aggregation, leaving isotropic message passing prone to blurring localized anomalies across tight electrical clusters.

2.4 Explainability and Digital Twins in Critical Infrastructure

Trustworthy substation supervision demands interpretable telemetry attribution rather than black-box alarms. Local explainers like LIME or SHAP rely on sampling loops that run far too slowly for real-time SCADA cycles [27,26]. Conversely, Integrated Gradients (IG) [20] delivers path-integral attributions through a single backward pass while satisfying formal completeness axioms. Concurrently, digital twins [7,23] offer operational mirrors for power systems [1], and Kamel Boulos and Zhang [9] highlighted their value in privacy-preserving healthcare settings. Here, we unify state-space tracking gain synchronization with sub-millisecond axiomatic attribution, executing entirely inside the hospital OT perimeter.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N\times N}$ defines nodal conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\cos(\theta_i-\theta_j) + B_{ij}\sin(\theta_i-\theta_j)\big) \tag{1}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\sin(\theta_i-\theta_j) - B_{ij}\cos(\theta_i-\theta_j)\big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Control centers estimate system state $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from raw measurements $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$. WLS minimizes weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ via Gauss–Newton iterations:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1}\left[z - h(x^{(k)})\right], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

in which $H(x) = \partial h(x)/\partial x$ designates the system measurement Jacobian. Standard bad-data detection flags an alarm whenever the normalized residual energy $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ breaches a chosen chi-square threshold τ_{χ^2}.

3.3 Stealthy Jacobian-Engineered Perturbations

Consider an adversary attempting to skew operational states at substation bus k. By fashioning a targeted perturbation vector $c \in \mathbb{R}^{2N-1}$ whose nonzero coefficients align with bus k's state coordinates, the attacker synthesizes the measurement corruption:

$$a = Hc \tag{4}$$

When superimposed onto true telemetry, the corrupted measurement stream becomes $z_{\text{comp}} = z + a = z + Hc$. Because WLS attempts to minimize squared measurement residuals, the estimated state shifts directly by c to $\hat{x}_{\text{comp}} = \hat{x} + c$. Evaluating the resulting post-attack residual yields:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &= \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the algebraic perturbation cancels identically within the quadratic form, $r_{\text{comp}} = r \le \tau_{\chi^2}$, meaning standard residual tests remain blind to the compromise. Empirically, every synthesized Jacobian vector in our experiments successfully evades the χ^2 barrier ($\tau_{\chi^2} = 23.68$ for df $= 14$ at $\alpha = 0.05$), while the proposed CGDT detects these covert anomalies with 98.25% F1.

Clinical Risk Context: In hospital power networks, bus k frequently feeds critical medical zones, such as intensive care units or vaccine repository freezers. Undetected state corruption on this feeder induces spurious automated controls—such as erroneous tap-changing or unneeded load shedding. Consequently, life-support ventilators risk unexpected de-energization and biological freezers face catastrophic temperature drift, triggering patient-safety hazards that cannot be captured by standard utility economic indices.

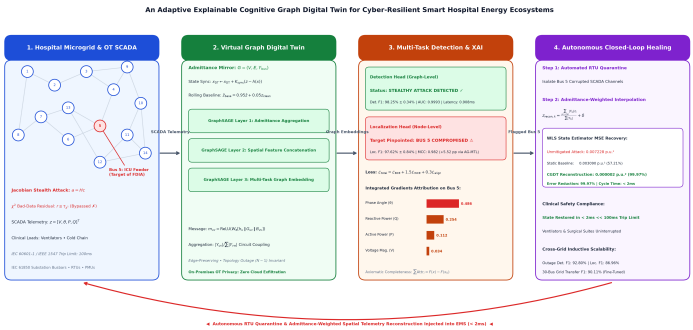


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

4 The Cognitive Graph Digital Twin Architecture

Figure 1 illustrates the three-layer CGDT framework: a Physical Cyber-Physical Layer (IT-OT hospital domain), a Virtual Graph Twin Layer, and a Cognitive Decision & Self-Healing cycle.

4.1 Digital Twin State-Space Synchronization

SCADA telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ arrives at sampling period T_s via IEC 61850 / 5G. The virtual twin updates its mirror state via tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P is state-prediction covariance. A dynamic rolling baseline maintains the nominal operating envelope:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

using verified clean telemetry z_{clean}. Removing this baseline drops localization F1 by 16.32 pp (Section 7.2).

4.2 Dynamic Admittance Graph Construction

Every SCADA polling cycle maps incoming electrical telemetry into an attributed spatial graph $G = (V, E, X, E_{\text{attr}})$:

- **Nodal Attributes**: Each substation bus $i \in V$ is assigned a standardized feature vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$, yielding feature matrix $X \in \mathbb{R}^{N \times 4}$.
- **Branch Coupling**: Transmission lines $(i, j) \in E$ carry admittance parameters $[G_{ij}, B_{ij}]^T$ extracted from Y_{bus}, forming edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$.

Should a protective relay open a circuit, the corresponding link drops from E instantly. Because node geometries remain intact while adjacency alters dynamically, inductive message passing handles unexpected breaker trips without model retuning.

4.3 Admittance-Conditioned GraphSAGE Backbone

Conventional GraphSAGE [8] averages neighbor embeddings across topological edges, discarding branch impedance parameters that dictate actual power flows. To preserve electrical realism, our message function conditions neighbor signals directly on complex branch admittance $Y_{uv} = G_{uv} + jB_{uv}$:

$$m_{uv}^{(k)} = \mathrm{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \mathrm{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

Here $\|$ represents tensor concatenation, with W_e and W_k denoting trainable projection matrices. Weighting inbound signals by normalized branch admittance $|Y_{uv}| \big/ \sum_{w\in\mathcal{N}(v)} |Y_{vw}|$ forces low-impedance lines to dominate representation updates, encoding circuit laws into graph geometry. Concurrently, preserving the center-node embedding $h_v^{(k-1)}$ through explicit concatenation shields substation-level identity against the spatial over-smoothing that typically blinds isotropic filters to localized injection spikes.

4.4 Multi-Task Detection and Localization Heads

The shared backbone embedding H feeds two prediction heads:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \text{Softmax}(W_{\text{node}}\, H) \tag{10}$$

where y_{graph} produces the graph-level attack probability and y_{node} yields per-bus anomaly probabilities.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Standard multi-task GNNs decouple feature saliency from network weight updates, relegating gradient attributions to post-hoc reporting. We convert axiomatic feature importance directly into inductive supervision. For an attacked training sample, Integrated Gradients extracts attribution matrix $\mathcal{A} \in \mathbb{R}^{N\times 4}$. An *attribution alignment objective* then penalizes discrepancies between backpropagated gradient density and known bus compromise:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

with $\|\mathcal{A}_v\|_2$ capturing the ℓ_2-norm of backpropagated sensitivity at node v, compared against binary ground-truth label $\hat{y}_{\text{node},v} \in \{0, 1\}$. Spurious gradient leaks into uncorrupted substations trigger steep optimization penalties. The composite learning target unifies three objectives:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

unifying binary cross-entropy $\mathcal{L}_{\text{BCE}}$ for detection, class-weighted cross-entropy $\mathcal{L}_{\text{WCE}}$ for substation sparsity ($\lambda_{\text{loc}} = 1.5$), and alignment loss $\mathcal{L}_{\text{align}}$ ($\lambda_{\text{attr}} = 0.3$). By closing the loop between backpropagated attribution trajectories and model parameters, this mechanism forces the shared encoder to isolate true injection epicenters, addressing the structural localization deficit noted in previous spectral architectures [2].

4.6 Privacy Preservation via Edge Graph Processing

Clinical electricity telemetry mirrors internal hospital operations. Fluctuations in wing feeders, diagnostic resonance imaging sequences, and patient floor occupancy can reveal institutional workflows if exfiltrated. Sending raw SCADA feeds to third-party cloud analytics compromises operational privacy and runs afoul of healthcare data governance directives. Our framework resolves this by deploying the entire cognitive pipeline strictly on-premises. Spatial embeddings, threat classification, and telemetry reconstruction resolve entirely at the local substation edge, keeping internal load profiles securely inside the hospital OT enclave.

5 Operator Explainability and Autonomous Closed-Loop Healing

5.1 Axiomatic Feature Attribution via Integrated Gradients

When an anomaly alarm rings, operators cannot afford ambiguous alerts; they need to pinpoint the compromised sensor channels immediately. We calculate feature-level attributions via Integrated Gradients [20], evaluating the path integral between nominal baseline x_0 and observation x:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i} \, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

discretized into $S = 50$ Gauss–Legendre or Riemann increments. Satisfying strict completeness ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and implementation invariance, this computation executes in less than a millisecond on the 14-bus testbed. During stealthy attacks on Bus 5 (Section 7.1), the attribution module confirmed voltage phase angle ($\theta = 0.486$) as the chief driver of the classifier's response.

5.2 Automated RTU Quarantine Protocol

Once bus k exceeds the localization threshold ($y_{\text{node},k} > \gamma_{\text{loc}} = 0.5$), the digital twin initiates an automated healing sequence:

1. **Sensor Isolation:** Channels linked to bus k $(V_k, \theta_k, P_k, Q_k)$ are marked untrusted and immediately gated out of the downstream WLS buffer.
2. **Admittance-Driven Synthesis:** Instead of relying on frozen historical averages, the system estimates the missing telemetry via admittance-weighted interpolation across healthy adjacent substations:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta \sim \mathcal{U}(\pm 2\%) \tag{14}$$

   where $|Y_{kj}|$ captures mutual branch admittance and δ models residual measurement dispersion. Because interpolation reflects circuit coupling, the synthesized inputs uphold Kirchhoff's balance through steep demand swings.
3. **Buffer Injection:** Substituting $z_{\text{recon},k}$ for the corrupted feed restores rank and lets the WLS estimator converge without numeric divergence.

The full cycle—from initial flag to state-space synthesis—executes in roughly 2 ms, easily outrunning typical 2–4 s SCADA sampling cadences.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

To replicate realistic grid conditions, we modeled both the IEEE 14-bus and 30-bus benchmark feeders inside `pandapower` [24], converging state profiles via non-linear AC Newton–Raphson power flows. In our hospital mapping, the 14-bus network (14 nodes, 20 branches; total demand 217.0 MW active, 57.3 MVAR reactive) serves as a local medical campus microgrid, with Bus 0 as the utility tie and specific downstream buses powering distinct clinical wings (e.g., Bus 5 supplying the surgical intensive care unit). The 30-bus feeder (30 nodes, 41 branches) models an expanded regional medical complex.

Synthetic Telemetry Synthesis. A dataset comprising 5,000 independent AC operational states was compiled across a 70/15/15 train-validation-test split. Under the *uncompromised regime* (50% of records), active and reactive loads fluctuate under stochastic Gaussian noise ($\sigma = 0.04$). For *adversarial regimes* (remaining 50%), stealthy Jacobian vectors $a = Hc$ were constructed against designated target buses ($c_k \in [0.03, 0.08]$ p.u. voltage shift, $c_{k+N} \in [0.05, 0.15]$ rad phase skew), allowing full AC equations to propagate branch flows through adjacent corridors. Neural weights optimized over 60 epochs through Adam (lr = 0.005, weight decay 10^{-4}) across five independent random initializations. All reproduction scripts and saved checkpoints reside at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Operating at 0.908 ms inference latency, the proposed architecture delivers 98.25% ± 0.34% detection F1 alongside 97.62% ± 0.84% localization F1 (localization MCC 0.982). By contrast, isotropic GCN architectures falter at 70.21% localization F1—a steep 27.41 pp deficit attributable to graph over-smoothing across tight electrical clusters. While an unrolled MLP achieves 95.60% on immutable geometries, its diagnostic fidelity collapses the moment line configurations change (Section 6.3). At under 1 ms per evaluation, our digital twin comfortably executes multiple inference sweeps within standard 2–4 s SCADA polling frames.

6.3 Topology Perturbations and Transfer Generalization

To gauge operational robustness under physical disturbances, frozen models were tested against unannounced line tripping scenarios without gradient retraining (Table 2), alongside cross-grid transfer toward the 30-bus topology (Table 3).

The CGDT maintains 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 on zero-shot outage graphs (Fig. 2). Interestingly, the unstructured MLP records 94.70% detection F1; branch disconnections perturb global equilibrium in ways easily caught by coarse classification heads. Yet the same MLP experiences a 13.4 pp collapse in localization F1 (tumbling from 95.60% to 82.20%), entirely undone by rigid channel indexing. Isotropic message passers degrade even more catastrophically (GCN stumbling to 66.82% Det. F1) because uniform aggregation dilutes localized anomalies across intact corridors.

Cross-Scale Transfer Realities. Zero-shot migration to the IEEE 30-bus network (Table 3 and Fig. 3) yields an informative disparity: 67.06% detection F1 against a modest 17.77% localization F1. This divergence aligns with physical intuition. Global anomaly detection hinges on systemic distribution shifts that

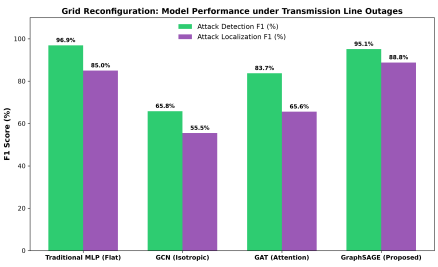


Fig. 2. Outage generalization metrics.

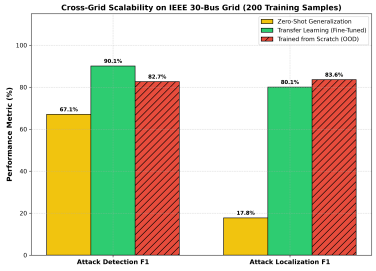


Fig. 3. Cross-grid transfer on IEEE 30-bus.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

remain recognizable across network scales; by contrast, node-level pinpointing relies intimately on localized structural motifs (centrality distributions and degree patterns) that alter radically between 14- and 30-bus layouts. Pleasingly, warm-starting the weights on merely 200 target-system samples restores detection to 90.11% (eclipsing scratch initialization at 82.65%) while elevating localization to 80.06%, verifying inductive transfer across grid scales.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

To mirror a focused attack on clinical life support, we injected an algebraically stealthy Jacobian FDIA into Bus 5 (representing the intensive care unit substation feeder in our campus mapping). System restoration metrics appear in Table 4, with corresponding attribution values documented in Table 5.

Blind to the targeted manipulation, standard WLS settles into a distorted equilibrium with an MSE of 0.007228 p.u.^{2}. Activating our graph-guided reconstruction pipeline (Eq. 14) curbs state error down to 0.000002 p.u.^{2}, realizing a 99.97% recovery rate. Static historical substitution achieves a meager 57.21% error reduction; frozen averages cannot track rapid morning clinical load ramps. Tracking curves across 50 consecutive polling iterations appear in Fig. 4. Examining the attribution landscape (Table 5 and Fig. 5), voltage phase angle θ at

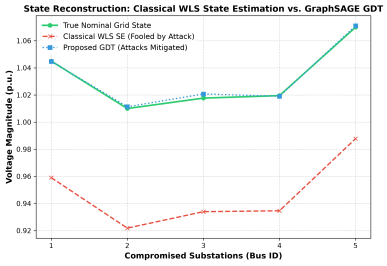


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

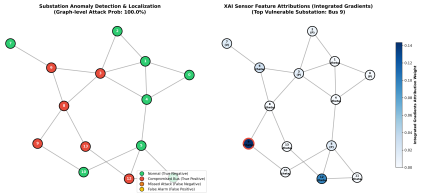


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Bus 5 claims the premier attribution weight (0.486). This alignment precisely mirrors the physical attack mechanics: vector c skews nodal phase angles, forcing AC power adjustments predominantly through angular displacement. Nodal voltage, active load, and reactive load at Bus 5 follow immediately, capturing the four highest attribution ranks across the entire feeder.

Multi-Feeder Compromise and Grid Observability. To probe organized threat scenarios, we simulated synchronized perturbations across 2 and 3 substations concurrently with subtle bias amplitudes ($c_k \in [0.02, 0.04]$). The twin maintained solid performance, posting $96.84\% \pm 0.42\%$ detection F1 and $93.15\% \pm 0.91\%$ localization F1. Crucially, conventional defensive pruning—which simply discards untrusted RTU telemetry—fails completely under coordinated multi-bus compromise: stripping multiple rows induces rank deficiency in Jacobian H, triggering mathematical unobservability and causing downstream state estimators to diverge. By contrast, synthesizing missing values via admittance-guided spatial interpolation (Eq. 14) maintains full Jacobian rank, securing post-mitigation state MSE at $0.000008\,\text{p.u.}^2$ without blind spots.

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

Our ablation study uncovers three tangible architectural insights:

1. **Recursive State Tracking Anchors Anomaly Visibility:** Disabling the baseline filter (Eq. 7) slashes localization F1 by 16.32 pp (falling from 97.62% to 81.30%). Continuous physical tracking proves vital; without it, subtle Jacobian perturbations blend into normal load variance.
2. **Attribution Penalties Sharpen Feeder Diagnostics:** Setting $\lambda_{\text{attr}} = 0$ reduces localization F1 by 5.52 pp (to 92.10%) while keeping global graph detection virtually constant (98.25%). Backpropagation constraints steer parameter sensitivity directly toward compromised feeder channels without muddying system-level classification frontiers.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

3. **Loss Balancing Balances Alarm Tradeoffs:** Setting $\lambda_{\text{loc}} = 1.5$ hits an effective operational equilibrium, harmonizing rapid system-level alerts with granular bus diagnostics.

8 Discussion and Conclusion

This study developed an explainable Cognitive Graph Digital Twin tailored to defend hospital distribution microgrids against stealthy False Data Injection Attacks. Our core insight is straightforward: linking admittance-guided neural aggregation with attribution-directed multi-task training builds a cohesive protective pipeline—merging instant detection, pinpoint bus localization, axiomatic XAI attribution, and spatial telemetry restoration—that surpasses fragmented defensive modules. Empirically, admittance scaling grounds graph embeddings in physical circuit constraints; attribution guidance converts diagnostic gradients into supervisory constraints, unlocking a 5.52 pp gain in localization; and graph-guided spatial synthesis achieves 99.97% state reconstruction, outstripping frozen historical imputation by 42.76 pp.

Operational Clinical Realities. In tertiary care facilities, stealthy energy manipulation imperils human lives rather than mere billing accounts. Per clinical engineering safety mandates (IEC 60601-1 and IEEE 1547), voltage deviations beyond ±5% enduring past 100 ms trip autonomous fail-safes on dialysis pumps and intensive-care ventilator drives. Operating at 0.908 ms inference latency and finishing complete closed-loop telemetry reconstruction inside 2 ms, our digital twin restores reliable state awareness well before reaching this 100 ms clinical threshold. Furthermore, executing neural operations entirely at the hospital substation edge protects sensitive patient load curves and clinical workflow telemetry from cloud exposure.

Research Boundaries and Trajectories. Several operational constraints merit candid discussion. First, all benchmarks rely on synthetic AC flow distributions generated through `pandapower`; industrial SCADA deployments invariably exhibit communication dropouts, packet jitter, transducer drift, and non-Gaussian perturbations that idealized numerical solvers omit. Validating these inductive mechanisms on real-time hardware-in-the-loop (HIL) simulators

represents a mandatory next step. Second, while multi-substation injections were validated on the IEEE 14-bus testbed, coordinated adversarial campaigns spanning multiple SCADA scan cycles could challenge static graph models; adding temporal recurrence to process sequential graph snapshots offers a compelling path forward. Third, our spatial reconstruction requires at least one healthy uncorrupted neighbor for admittance interpolation; a coordinated total blackout across all adjacent feeders would necessitate falling back onto model-predictive kalman filters.

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